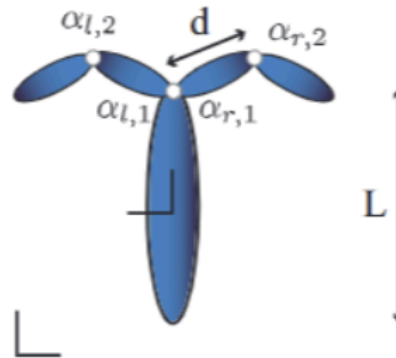


Problem 1

1 Gait optimization for a flagellar swimmer



Develop the equations of motion and simulate the locomotion of a flagellar based swimmer as shown above. The center link is L long with a floating base reference frame attached at the center of the link. There are two appendages on the left and right side. Each appendage has two joints and two links that are controllable. Set the joint angles such that when all angles are zero the limbs point straight up along the length of the body. Define the angle directions such that a positive increase in the left joints results in a CCW rotation, and a positive increase in the right joints results in a CW rotation. Thus setting $\alpha_{r,i}(t) = \alpha_{l,i}(t)$ results in symmetric motion of the limbs across the left and right side of the body. If we restrict ourselves to symmetric gaits then the shape space is two dimensional $\alpha_1(t) = \alpha_{r,1}(t) = \alpha_{l,1}(t)$ and $\alpha_2(t) = \alpha_{r,2}(t) = \alpha_{l,2}(t)$.

Part 1

Design by hand a gait that will move the robot forward

$\alpha_{l,1} = \alpha_{l,1} = \alpha_1$, $\alpha_{l,2} = \alpha_{l,2} = \alpha_2$ α_1 (controls the inner joints), α_2 (controls the outer joints)
 At low Reynolds number we need a closed loop in shape space. Let $\alpha_1(t) = A \sin(2\pi t)$, $\alpha_2(t) = A \sin(2\pi t + \phi)$ let $\phi = \frac{\pi}{2}$
 $\Rightarrow d_1(t) = A \sin(2\pi t)$, $d_2(t) = A \cos(2\pi t)$ this will create a circular/elliptical loop in (d_1, d_2) space let $A = \frac{\pi}{4}$
 $\Rightarrow \alpha_1(t) = \frac{\pi}{4} \sin(2\pi t)$ & $\alpha_2(t) = \frac{\pi}{4} \cos(2\pi t) \Rightarrow$ This gait moves the inner and outer joints out of phase which creates different drag forces during different parts of the cycle. This gait is symmetric, periodic, non-reciprocal gait with a phase shift of 90°

```
In [1]: import sympy as sp
import numpy as np
from sympy import lambdify, cos, sin, Matrix, simplify

# Helper Functions
def cross_matrix(vec):

    return np.array([[0, -vec[2,0], vec[1,0]],
                    [vec[2,0], 0, -vec[0,0]],
                    [-vec[1,0], vec[0,0], 0]])

def uncross_matrix(mat):

    return np.array([[mat[2,1]],
                    [mat[0,2]],
```

```

        [-mat[0,1]]])

def twist2vec(twist):
    m = Matrix.zeros(6,1)

    m[0:3,0] = uncross_matrix(twist[0:3,0:3])
    m[3:,0] = twist[0:3,3]

    return m

def R_z(theta):
    return np.array([[cos(theta), -sin(theta), 0],
                     [sin(theta), cos(theta), 0],
                     [0, 0, 1]])

def R_y(theta):
    return np.array([[cos(theta), 0, sin(theta)],
                     [0, 1, 0],
                     [-sin(theta), 0, cos(theta)]])

def R_x(theta):
    return np.array([[1, 0, 0],
                     [0, cos(theta), -sin(theta)],
                     [0, sin(theta), cos(theta)]])

def T(R, p):
    m = Matrix.zeros(4,4)
    m[0:3, 0:3] = R
    m[:3, 3] = p
    m[3, 3] = 1
    return m

# Geometry
t = sp.symbols("t", real=True)

L = sp.symbols("L", real=True) # body Length
d = sp.symbols("d", real=True) # appendage Link Length
k = sp.symbols("k", real=True) # drag coefficient

# Configuration variables
x = sp.Function('x', real=True)(t)
y = sp.Function('y', real=True)(t)
theta = sp.Function('theta', real=True)(t)

alpha1 = sp.Function('alpha1', real=True)(t)
alpha2 = sp.Function('alpha2', real=True)(t)

# q = pose + symmetric shape variables
q = [x, y, theta, alpha1, alpha2]
q_dt = [sp.diff(qq, t) for qq in q]
q_dtdt = [sp.diff(qq, t, t) for qq in q]

# Transformation matrices
# World to body center
T_ob = T(R_z(theta), Matrix([[x], [y], [0]]))

```

```

# Body COM
T_b = T_0b

# Appendage joint location at front of body
T_body_to_joint = T(Matrix.eye(3), Matrix([[L/2], [0], [0]]))

# Left link 1 COM
T_l1 = T_0b @ T_body_to_joint @ T(R_z(alpha1), Matrix([[0], [0], [0]])) @ T(Matrix.

# Left link 2 COM
T_l2 = T_0b @ T_body_to_joint @ T(R_z(alpha1), Matrix([[0], [0], [0]])) @ T(Matrix.

# Right link 1 COM
# Right side uses negative angle so positive alpha bends clockwise
T_r1 = T_0b @ T_body_to_joint @ T(R_z(-alpha1), Matrix([[0], [0], [0]])) @ T(Matrix

# Right link 2 COM
T_r2 = T_0b @ T_body_to_joint @ T(R_z(-alpha1), Matrix([[0], [0], [0]])) @ T(Matri

# Body twists
# Vb = T^-1 * Tdot
body_twist_b = T_b.inv() @ sp.diff(T_b, t)
body_twist_l1 = T_l1.inv() @ sp.diff(T_l1, t)
body_twist_l2 = T_l2.inv() @ sp.diff(T_l2, t)
body_twist_r1 = T_r1.inv() @ sp.diff(T_r1, t)
body_twist_r2 = T_r2.inv() @ sp.diff(T_r2, t)

# Twist vectors
v_b = twist2vec(body_twist_b)
v_l1 = twist2vec(body_twist_l1)
v_l2 = twist2vec(body_twist_l2)
v_r1 = twist2vec(body_twist_r1)
v_r2 = twist2vec(body_twist_r2)

# Body Jacobians
J_b = v_b.jacobian(q_dt)
J_l1 = v_l1.jacobian(q_dt)
J_l2 = v_l2.jacobian(q_dt)
J_r1 = v_r1.jacobian(q_dt)
J_r2 = v_r2.jacobian(q_dt)

# Drag matrices
# Body drag matrix, body length L
B_body = -Matrix([
    [0, 0, 0, 0, 0, 0],
    [0, 0, 0, 0, 0, 0],
    [0, 0, 2/3*k*L**3, 0, 0, 0],
    [0, 0, 0, k*L, 0, 0],
    [0, 0, 0, 0, 2*k*L, 0],
    [0, 0, 0, 0, 0, 0]
])

# Appendage Link drag matrix, Link length d
B_link = -Matrix([

```

```

[0, 0, 0,          0, 0, 0],
[0, 0, 0,          0, 0, 0],
[0, 0, 2/3*k*d**3, 0, 0, 0],
[0, 0, 0,          k*d, 0, 0],
[0, 0, 0,          0, 2*k*d, 0],
[0, 0, 0,          0, 0, 0]
])

# Generalized force balance matrix
A = J_b.T @ B_body @ J_b + J_l1.T @ B_link @ J_l1 + J_l2.T @ B_link @ J_l2 + J_r1.

A_11 = A[0:3, 0:3]
A_12 = A[0:3, 3:]
A_21 = A[3:, 0:3]
A_22 = A[3:, 3:]

# Local connection matrix
# pose_dot = -A_11^-1 A_12 shape_dot
A_tilde = -A_11.LUsolve(A_12)

# Lambdify
params = [L, d, k]

A_11_lambda = lambdify([t, q, q_dt, params], A_11, "numpy")
A_12_lambda = lambdify([t, q, q_dt, params], A_12, "numpy")
A_tilde_lambda = lambdify([t, q, q_dt, params], A_tilde, "numpy")

def connection_matrix(t_num, q_num, q_dt_num, params_num):
    A11 = A_11_lambda(t_num, q_num, q_dt_num, params_num)
    A12 = A_12_lambda(t_num, q_num, q_dt_num, params_num)
    return -np.linalg.solve(A11, A12)

```

```

In [2]: # To check the zero-angle configuration it should only have a x-direction and zero
test_subs = {
    x: 0,
    y: 0,
    theta: 0,
    alpha1: 0,
    alpha2: 0,
    L: 2,
    d: 1
}

simplify(T_b.subs(test_subs)), simplify(T_l1.subs(test_subs)), simplify(T_l2.subs(t

```

```
Out[2]: (Matrix([
  [1, 0, 0, 0],
  [0, 1, 0, 0],
  [0, 0, 1, 0],
  [0, 0, 0, 1]]),
Matrix([
  [1, 0, 0, 3/2],
  [0, 1, 0, 0],
  [0, 0, 1, 0],
  [0, 0, 0, 1]]),
Matrix([
  [1, 0, 0, 5/2],
  [0, 1, 0, 0],
  [0, 0, 1, 0],
  [0, 0, 0, 1]]),
Matrix([
  [1, 0, 0, 3/2],
  [0, 1, 0, 0],
  [0, 0, 1, 0],
  [0, 0, 0, 1]]),
Matrix([
  [1, 0, 0, 5/2],
  [0, 1, 0, 0],
  [0, 0, 1, 0],
  [0, 0, 0, 1]]))
```

```
In [3]: import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

# Part 1: hand-designed gait
# alpha1 = pi/4 sin(2 pi t)
# alpha2 = pi/4 cos(2 pi t)
A_gait = np.pi / 4

def gait(t_num):
    alpha1 = A_gait * np.sin(2*np.pi*t_num)
    alpha2 = A_gait * np.cos(2*np.pi*t_num)

    alpha1_dot = A_gait * 2*np.pi * np.cos(2*np.pi*t_num)
    alpha2_dot = -A_gait * 2*np.pi * np.sin(2*np.pi*t_num)

    return alpha1, alpha2, alpha1_dot, alpha2_dot

# Swimmer dynamics
# z = [x, y, theta]
def swimmer_dynamics(t_num, z, params_num):
    x_num, y_num, theta_num = z

    alpha1, alpha2, alpha1_dot, alpha2_dot = gait(t_num)

    q_num = np.array([
        x_num,
        y_num,
        theta_num,
        alpha1,
```

```

        alpha2
    ], dtype=float)

    q_dot_num = np.array([
        0,
        0,
        0,
        alpha1_dot,
        alpha2_dot
    ], dtype=float)

    A_tilde_num = np.array(
        A_tilde_lambda(t_num, q_num, q_dot_num, params_num),
        dtype=float
    )

    shape_dot = np.array([
        alpha1_dot,
        alpha2_dot
    ], dtype=float)

    pose_dot = A_tilde_num @ shape_dot

    return pose_dot.flatten()

# Simulate one gait cycle
params_num = [1.0, 0.5, 1.0] # [L, d, k]

z0 = np.array([0.0, 0.0, 0.0])

t_span = (0.0, 1.0)
time = np.linspace(0.0, 1.0, 500)

sol = solve_ivp(
    fun=lambda t_num, z: swimmer_dynamics(t_num, z, params_num),
    t_span=t_span,
    y0=z0,
    t_eval=time,
    rtol=1e-8,
    atol=1e-8
)

x_sol = sol.y[0]
y_sol = sol.y[1]
theta_sol = sol.y[2]

alpha_1_sol = np.zeros_like(time)
alpha_2_sol = np.zeros_like(time)
alpha_1_dot_sol = np.zeros_like(time)
alpha_2_dot_sol = np.zeros_like(time)

x_dot_sol = np.zeros_like(time)
y_dot_sol = np.zeros_like(time)
theta_dot_sol = np.zeros_like(time)

for i, ti in enumerate(time):

```

```

alpha_1_sol[i], alpha_2_sol[i], alpha_1_dot_sol[i], alpha_2_dot_sol[i] = gait(t

zdot = swimmer_dynamics(
    ti,
    np.array([x_sol[i], y_sol[i], theta_sol[i]]),
    params_num
)

x_dot_sol[i] = zdot[0]
y_dot_sol[i] = zdot[1]
theta_dot_sol[i] = zdot[2]

# Print net motion
print("Net x displacement:", x_sol[-1] - x_sol[0])
print("Net y displacement:", y_sol[-1] - y_sol[0])
print("Net theta change:", theta_sol[-1] - theta_sol[0])

fig = plt.figure(1, figsize=(10, 10))
fig.clf()
ax = fig.subplots(5, 2, sharex=True)

ax[0][0].plot(time, x_sol)
ax[0][0].set_ylabel('x')

ax[1][0].plot(time, y_sol)
ax[1][0].set_ylabel('y')

ax[2][0].plot(time, theta_sol)
ax[2][0].set_ylabel('theta')

ax[3][0].plot(time, alpha_1_sol)
ax[3][0].set_ylabel('alpha_1')

ax[4][0].plot(time, alpha_2_sol)
ax[4][0].set_ylabel('alpha_2')
ax[4][0].set_xlabel('time')

ax[0][1].plot(time, x_dot_sol)
ax[0][1].set_ylabel('x_dot')

ax[1][1].plot(time, y_dot_sol)
ax[1][1].set_ylabel('y_dot')

ax[2][1].plot(time, theta_dot_sol)
ax[2][1].set_ylabel('theta_dot')

ax[3][1].plot(time, alpha_1_dot_sol)
ax[3][1].set_ylabel('alpha_dot_1')

ax[4][1].plot(time, alpha_2_dot_sol)
ax[4][1].set_ylabel('alpha_dot_2')
ax[4][1].set_xlabel('time')

plt.tight_layout()

```

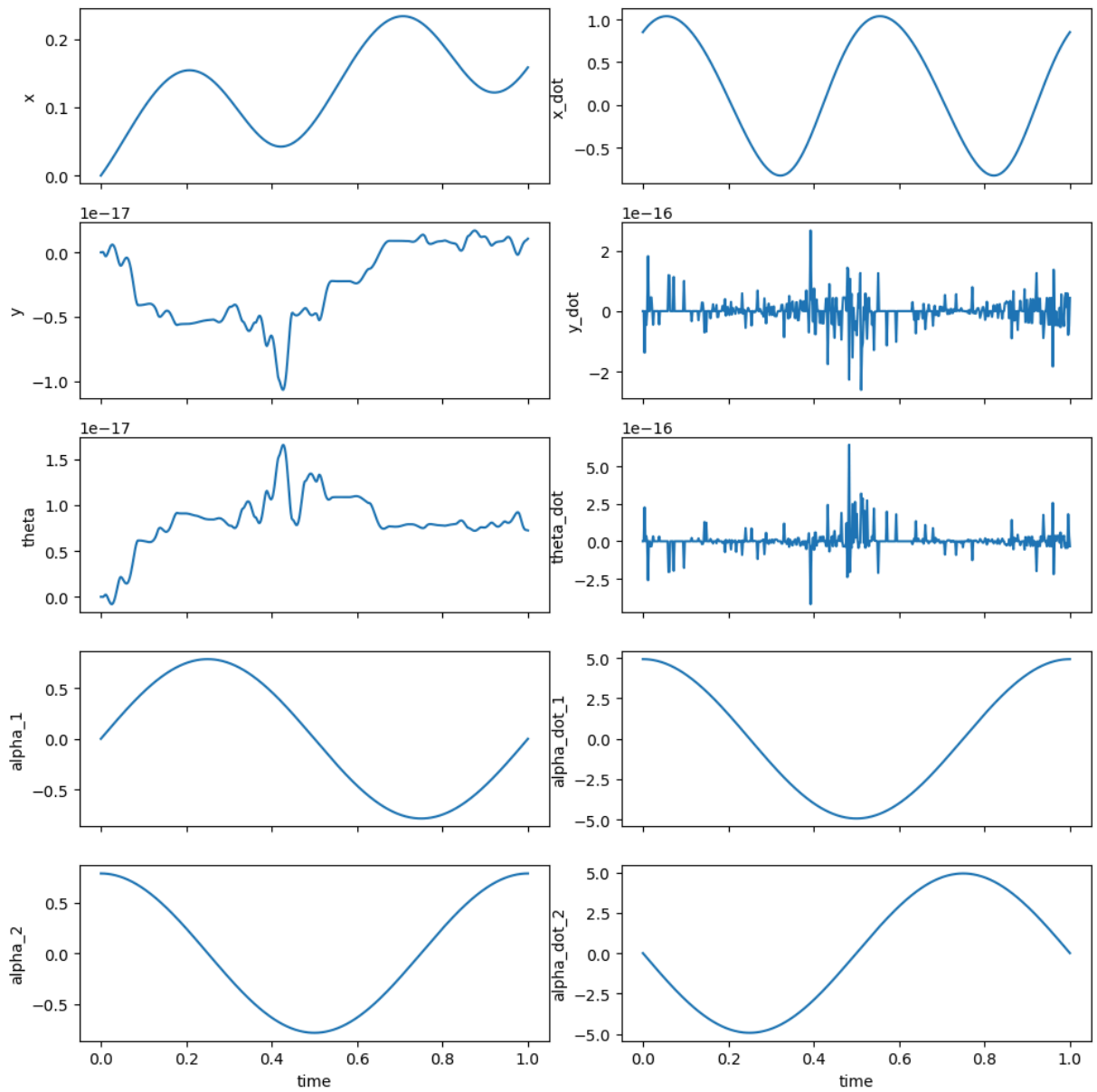
```

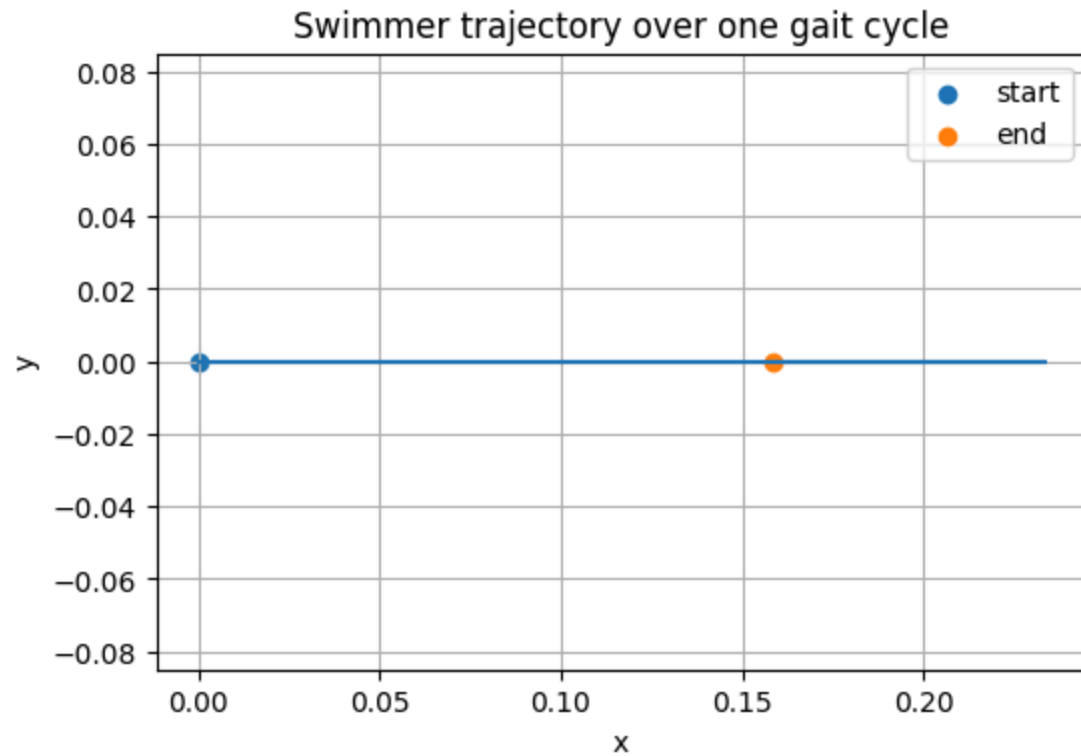
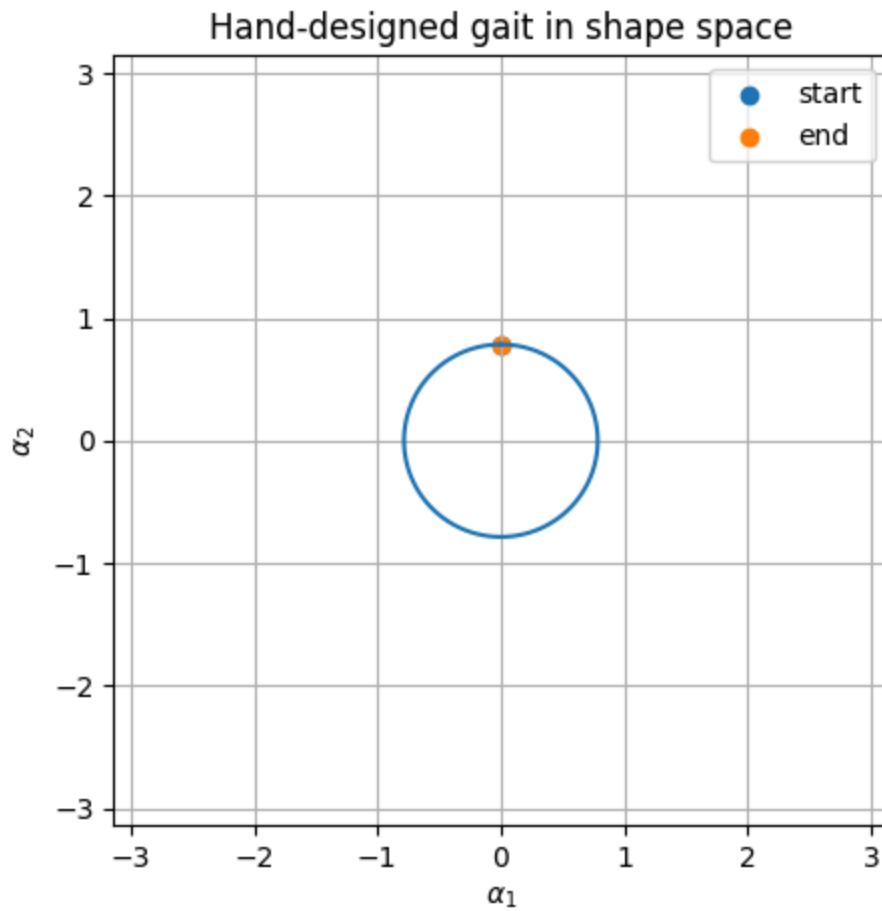
# Shape-space gait plot
plt.figure(10, figsize=(5, 5))
plt.plot(alpha_1_sol, alpha_2_sol)
plt.scatter(alpha_1_sol[0], alpha_2_sol[0], label='start')
plt.scatter(alpha_1_sol[-1], alpha_2_sol[-1], label='end')
plt.gca().axis([-np.pi, np.pi, -np.pi, np.pi])
plt.xlabel(r'$\alpha_1$')
plt.ylabel(r'$\alpha_2$')
plt.title('Hand-designed gait in shape space')
plt.grid(True)
plt.legend()

plt.figure(11, figsize=(6, 4))
plt.plot(x_sol, y_sol)
plt.scatter(x_sol[0], y_sol[0], label='start')
plt.scatter(x_sol[-1], y_sol[-1], label='end')
plt.xlabel('x')
plt.ylabel('y')
plt.title('Swimmer trajectory over one gait cycle')
plt.axis('equal')
plt.grid(True)
plt.legend()
plt.show()

```

Net x displacement: 0.15838455685340302
 Net y displacement: 1.0642684725136018e-18
 Net theta change: 7.239625399882056e-18





Part 2

Develop a gait generation function using a truncated Fourier series for α_1, α_2

```
In [4]: def fourier_gait(t_num, coeffs, omega=2*np.pi):
        """
        Truncated Fourier series gait for alpha1 and alpha2.

        coeffs format:
        coeffs = {
            "a1_cos": [...],
            "a1_sin": [...],
            "a2_cos": [...],
            "a2_sin": [...],
        }

        Each list has length N.
        """

        a1_cos = np.array(coeffs["a1_cos"])
        a1_sin = np.array(coeffs["a1_sin"])
        a2_cos = np.array(coeffs["a2_cos"])
        a2_sin = np.array(coeffs["a2_sin"])

        N = len(a1_cos)

        alpha1 = 0.0
        alpha2 = 0.0
        alpha1_dot = 0.0
        alpha2_dot = 0.0

        for k in range(1, N + 1):
            alpha1 += a1_cos[k-1]*np.cos(k*omega*t_num) + a1_sin[k-1]*np.sin(k*omega*t_num)
            alpha2 += a2_cos[k-1]*np.cos(k*omega*t_num) + a2_sin[k-1]*np.sin(k*omega*t_num)

            alpha1_dot += -a1_cos[k-1]*k*omega*np.sin(k*omega*t_num) + a1_sin[k-1]*k*omega*np.cos(k*omega*t_num)
            alpha2_dot += -a2_cos[k-1]*k*omega*np.sin(k*omega*t_num) + a2_sin[k-1]*k*omega*np.cos(k*omega*t_num)

        return alpha1, alpha2, alpha1_dot, alpha2_dot
```

```
In [5]: coeffs_test = {
        "a1_cos": [0.0],
        "a1_sin": [np.pi/4],
        "a2_cos": [np.pi/4],
        "a2_sin": [0.0]
    }

    time = np.linspace(0, 1, 500)

    alpha_1_sol = np.zeros_like(time)
    alpha_2_sol = np.zeros_like(time)
    alpha_1_dot_sol = np.zeros_like(time)
    alpha_2_dot_sol = np.zeros_like(time)

    for i, ti in enumerate(time):
        alpha_1_sol[i], alpha_2_sol[i], alpha_1_dot_sol[i], alpha_2_dot_sol[i] = fourier_gait(ti, coeffs_test)
```

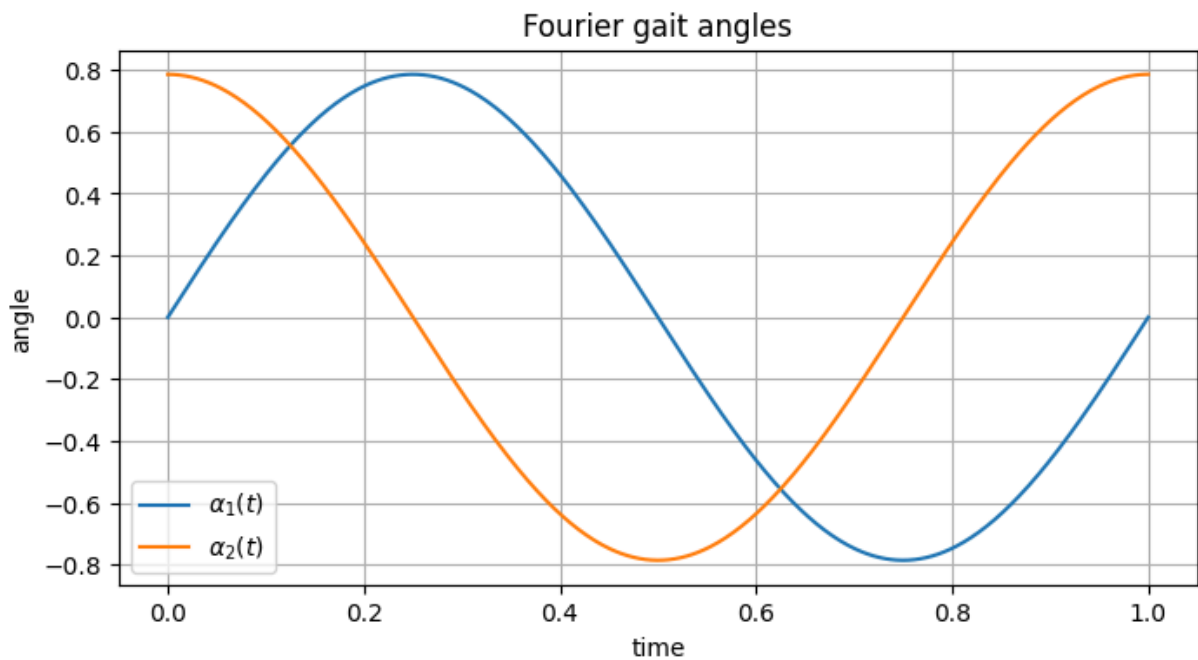
```

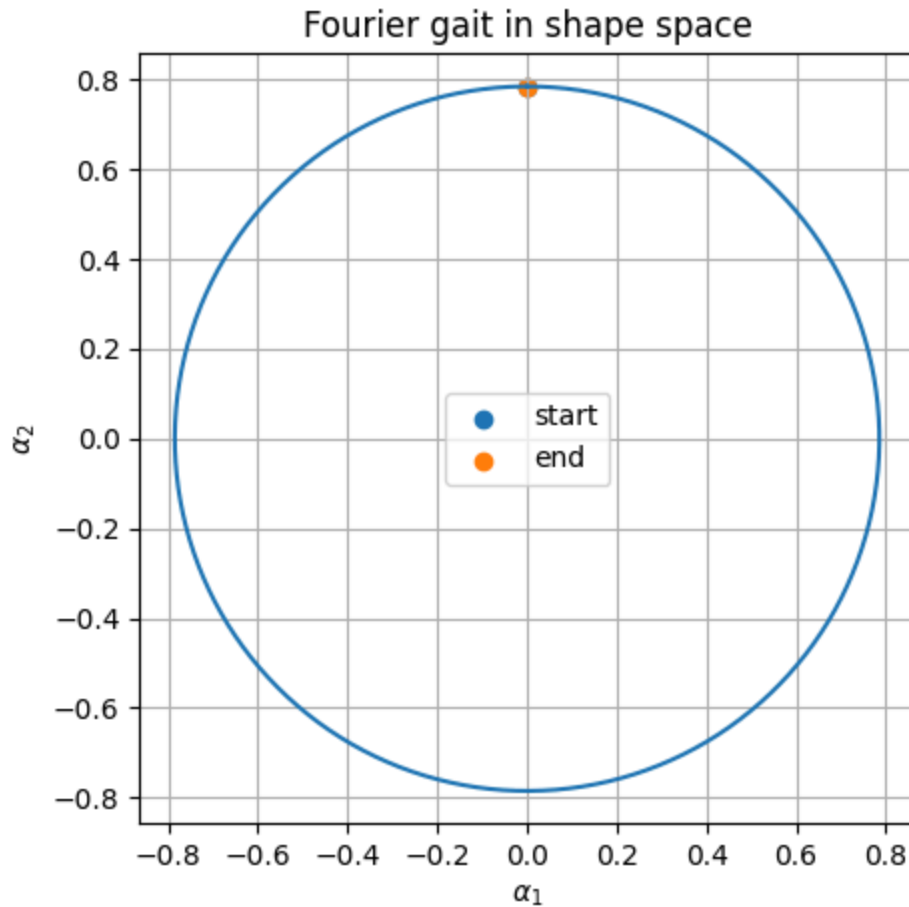
        coeffs_test
    )

plt.figure(figsize=(8, 4))
plt.plot(time, alpha_1_sol, label=r'$\alpha_1(t)$')
plt.plot(time, alpha_2_sol, label=r'$\alpha_2(t)$')
plt.xlabel('time')
plt.ylabel('angle')
plt.title('Fourier gait angles')
plt.grid(True)
plt.legend()
plt.show()

plt.figure(figsize=(5, 5))
plt.plot(alpha_1_sol, alpha_2_sol)
plt.scatter(alpha_1_sol[0], alpha_2_sol[0], label='start')
plt.scatter(alpha_1_sol[-1], alpha_2_sol[-1], label='end')
plt.xlabel(r'$\alpha_1$')
plt.ylabel(r'$\alpha_2$')
plt.title('Fourier gait in shape space')
plt.axis('equal')
plt.grid(True)
plt.legend()
plt.show()

```





```
In [7]: import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import minimize

# Optimization for maximum forward distance
N_fourier = 1
params_num = [1.0, 0.5, 1.0] # [L, d, k]

dt = 1/60
time_opt = np.arange(0.0, 1.0 + dt, dt)

z0 = np.array([0.0, 0.0, 0.0])

# Fourier gait with offsets
# c = [alpha1_offset, alpha2_offset,
#      alpha1_cos, alpha1_sin,
#      alpha2_cos, alpha2_sin]
def fourier_gait_with_offset(t_num, c, omega=2*np.pi):
    a10 = c[0]
    a20 = c[1]

    a1_cos = c[2]
    a1_sin = c[3]
    a2_cos = c[4]
    a2_sin = c[5]
```

```

alpha1 = a10 + a1_cos*np.cos(omega*t_num) + a1_sin*np.sin(omega*t_num)
alpha2 = a20 + a2_cos*np.cos(omega*t_num) + a2_sin*np.sin(omega*t_num)

alpha1_dot = -a1_cos*omega*np.sin(omega*t_num) + a1_sin*omega*np.cos(omega*t_num)
alpha2_dot = -a2_cos*omega*np.sin(omega*t_num) + a2_sin*omega*np.cos(omega*t_num)

return alpha1, alpha2, alpha1_dot, alpha2_dot

# Dynamics using Fourier gait
def swimmer_dynamics_fourier_offset(t_num, z, c):
    x_num, y_num, theta_num = z

    alpha1, alpha2, alpha1_dot, alpha2_dot = fourier_gait_with_offset(t_num, c)

    q_num = np.array([
        x_num,
        y_num,
        theta_num,
        alpha1,
        alpha2
    ], dtype=float)

    q_dot_num = np.array([
        0.0,
        0.0,
        0.0,
        alpha1_dot,
        alpha2_dot
    ], dtype=float)

    A_tilde_num = np.array(
        A_tilde_lambda(t_num, q_num, q_dot_num, params_num),
        dtype=float
    )

    shape_dot = np.array([
        alpha1_dot,
        alpha2_dot
    ], dtype=float)

    pose_dot = A_tilde_num @ shape_dot

    return pose_dot.flatten()

# Euler simulation over one cycle
def simulate_one_cycle(c, dt=1/60):
    z = z0.copy()

    x_hist = []
    y_hist = []
    theta_hist = []

    alpha1_hist = []
    alpha2_hist = []
    alpha1dot_hist = []
    alpha2dot_hist = []

```

```

xdot_hist = []
ydot_hist = []
thetadot_hist = []

time_hist = np.arange(0.0, 1.0 + dt, dt)

for ti in time_hist:
    alpha1, alpha2, alpha1_dot, alpha2_dot = fourier_gait_with_offset(ti, c)

    zdot = swimmer_dynamics_fourier_offset(ti, z, c)

    x_hist.append(z[0])
    y_hist.append(z[1])
    theta_hist.append(z[2])

    alpha1_hist.append(alpha1)
    alpha2_hist.append(alpha2)
    alpha1dot_hist.append(alpha1_dot)
    alpha2dot_hist.append(alpha2_dot)

    xdot_hist.append(zdot[0])
    ydot_hist.append(zdot[1])
    thetadot_hist.append(zdot[2])

    z = z + dt * zdot

    if not np.all(np.isfinite(z)):
        break

return {
    "time": np.array(time_hist[:len(x_hist)]),
    "x": np.array(x_hist),
    "y": np.array(y_hist),
    "theta": np.array(theta_hist),
    "alpha1": np.array(alpha1_hist),
    "alpha2": np.array(alpha2_hist),
    "alpha1_dot": np.array(alpha1dot_hist),
    "alpha2_dot": np.array(alpha2dot_hist),
    "x_dot": np.array(xdot_hist),
    "y_dot": np.array(ydot_hist),
    "theta_dot": np.array(thetadot_hist),
}

# Objective: maximize forward x distance
def objective_max_forward(c):
    try:
        sim = simulate_one_cycle(c, dt=dt)

        if len(sim["x"]) < 5:
            return 1e6

        net_x = sim["x"][-1] - sim["x"][0]

        if not np.isfinite(net_x):
            return 1e6

```

```

        return -net_x

    except Exception:
        return 1e6

# Initial guess
# alpha1 = 0.6 + pi/6 sin(2pi t)
# alpha2 = 0.6 + pi/6 cos(2pi t)
c0 = np.array([
    0.6,          # alpha1 offset
    0.6,          # alpha2 offset
    0.0,          # alpha1 cos
    np.pi/6,     # alpha1 sin
    np.pi/6,     # alpha2 cos
    0.0           # alpha2 sin
], dtype=float)

# Bounds to make optimization faster and avoid singularities
bounds = [
    (0.35, 1.2),  # alpha1 offset
    (0.35, 1.2),  # alpha2 offset
    (-1.0, 1.0),  # alpha1 cos
    (-1.0, 1.0),  # alpha1 sin
    (-1.0, 1.0),  # alpha2 cos
    (-1.0, 1.0),  # alpha2 sin
]

# Run optimizer
result_forward = minimize(
    objective_max_forward,
    c0,
    method="SLSQP",
    bounds=bounds,
    options={
        "maxiter": 20,
        "ftol": 1e-4,
        "disp": True
    }
)

c_opt_forward = result_forward.x

print("Optimization success:", result_forward.success)
print("Optimizer message:", result_forward.message)
print("Maximum forward distance:", -result_forward.fun)
print("Optimized coefficients:")
print(c_opt_forward)

# Simulate optimized gait
sim = simulate_one_cycle(c_opt_forward, dt=dt)

time = sim["time"]
x_sol = sim["x"]
y_sol = sim["y"]
theta_sol = sim["theta"]

```



```

alpha_1_sol = sim["alpha1"]
alpha_2_sol = sim["alpha2"]

alpha_1_dot_sol = sim["alpha1_dot"]
alpha_2_dot_sol = sim["alpha2_dot"]

x_dot_sol = sim["x_dot"]
y_dot_sol = sim["y_dot"]
theta_dot_sol = sim["theta_dot"]

print("Net x displacement:", x_sol[-1] - x_sol[0])
print("Net y displacement:", y_sol[-1] - y_sol[0])
print("Net theta change:", theta_sol[-1] - theta_sol[0])

# Plotting states and velocities
fig = plt.figure(1, figsize=(11, 10))
fig.clf()
ax = fig.subplots(5, 2, sharex=True)

ax[0][0].plot(time, x_sol)
ax[0][0].set_ylabel('x')

ax[1][0].plot(time, y_sol)
ax[1][0].set_ylabel('y')

ax[2][0].plot(time, theta_sol)
ax[2][0].set_ylabel('theta')

ax[3][0].plot(time, alpha_1_sol)
ax[3][0].set_ylabel('alpha_1')

ax[4][0].plot(time, alpha_2_sol)
ax[4][0].set_ylabel('alpha_2')
ax[4][0].set_xlabel('time')

ax[0][1].plot(time, x_dot_sol)
ax[0][1].set_ylabel('x_dot')

ax[1][1].plot(time, y_dot_sol)
ax[1][1].set_ylabel('y_dot')

ax[2][1].plot(time, theta_dot_sol)
ax[2][1].set_ylabel('theta_dot')

ax[3][1].plot(time, alpha_1_dot_sol)
ax[3][1].set_ylabel('alpha_dot_1')

ax[4][1].plot(time, alpha_2_dot_sol)
ax[4][1].set_ylabel('alpha_dot_2')
ax[4][1].set_xlabel('time')

plt.tight_layout()

# Shape-space plot

```

```

plt.figure(2, figsize=(5, 5))
plt.plot(alpha_1_sol, alpha_2_sol)
plt.scatter(alpha_1_sol[0], alpha_2_sol[0], label='start')
plt.scatter(alpha_1_sol[-1], alpha_2_sol[-1], label='end')
plt.xlabel(r'$\alpha_1$')
plt.ylabel(r'$\alpha_2$')
plt.title('Optimized gait in shape space')
plt.grid(True)
plt.axis('equal')
plt.legend()

# XY trajectory plot
plt.figure(3, figsize=(6, 4))
plt.plot(x_sol, y_sol)
plt.scatter(x_sol[0], y_sol[0], label='start')
plt.scatter(x_sol[-1], y_sol[-1], label='end')
plt.xlabel('x')
plt.ylabel('y')
plt.title('Optimized swimmer trajectory')
plt.grid(True)
plt.axis('equal')
plt.legend()

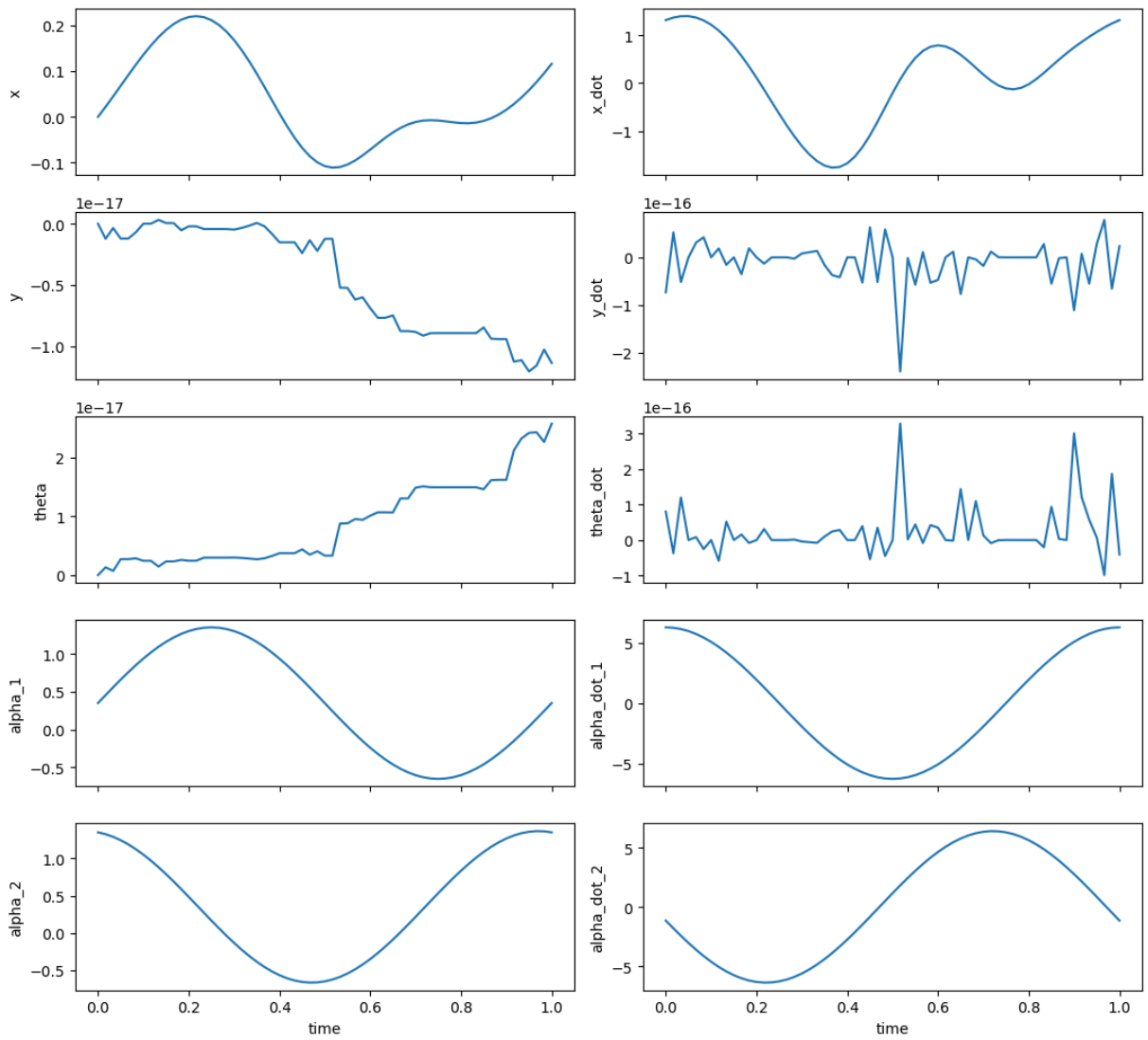
plt.show()

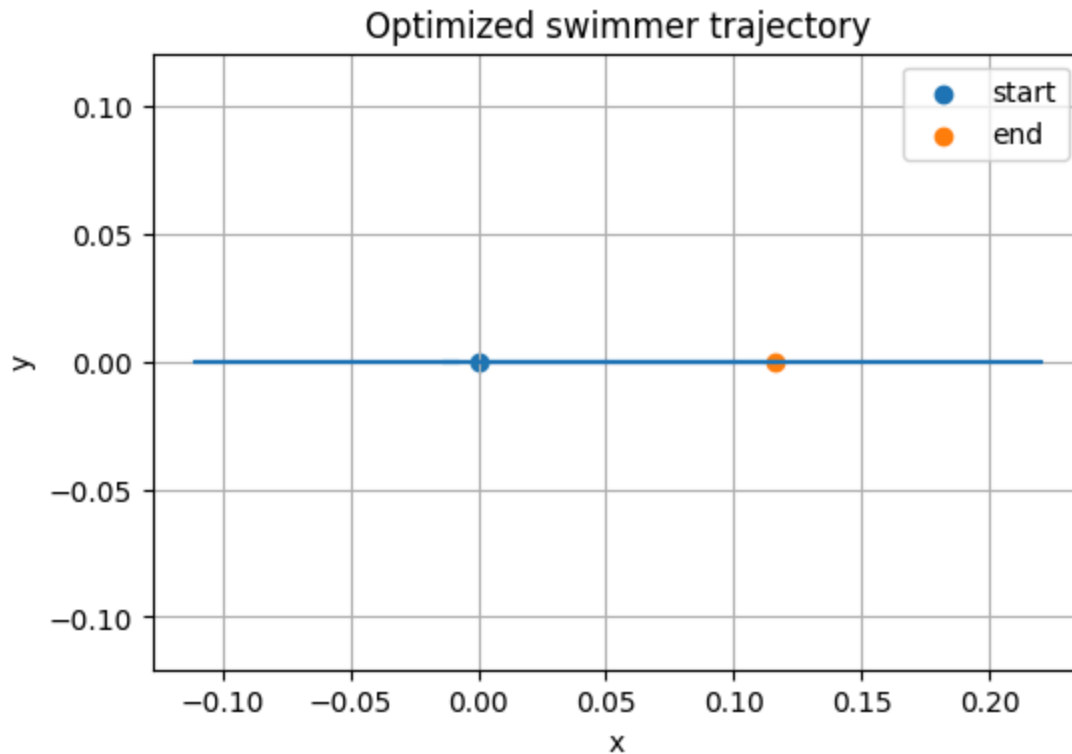
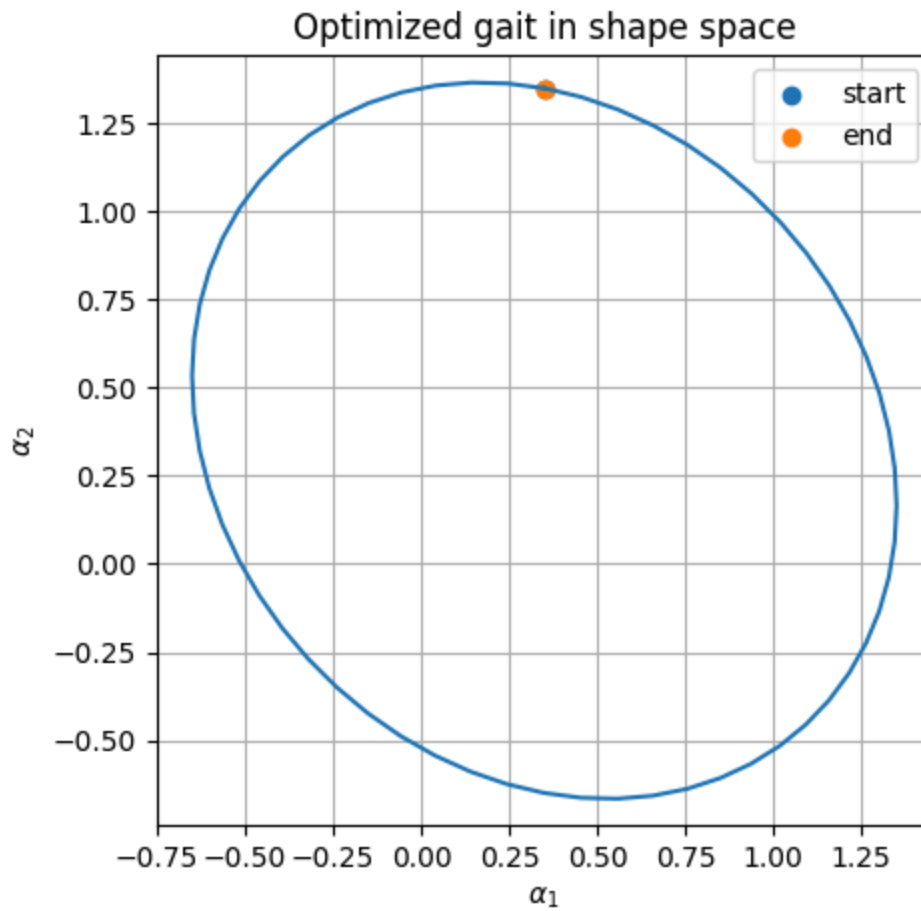
```

```

Optimization terminated successfully    (Exit mode 0)
      Current function value: -0.11598015258942182
      Iterations: 10
      Function evaluations: 70
      Gradient evaluations: 10
Optimization success: True
Optimizer message: Optimization terminated successfully
Maximum forward distance: 0.11598015258942182
Optimized coefficients:
[ 0.35      0.35      0.001185    1.          1.         -0.18419131]
Net x displacement: 0.11598015258942182
Net y displacement: -1.1389392559608361e-17
Net theta change: 2.5735390534280784e-17

```





```
In [8]: import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import minimize
```

```

# Minimum COT optimization
dt = 1/60
params_num = [1.0, 0.5, 1.0] # [L, d, k]
z0 = np.array([0.0, 0.0, 0.0])

# Fourier gait with offsets
def fourier_gait_with_offset(t_num, c, omega=2*np.pi):
    a10 = c[0]
    a20 = c[1]

    a1_cos = c[2]
    a1_sin = c[3]
    a2_cos = c[4]
    a2_sin = c[5]

    alpha1 = a10 + a1_cos*np.cos(omega*t_num) + a1_sin*np.sin(omega*t_num)
    alpha2 = a20 + a2_cos*np.cos(omega*t_num) + a2_sin*np.sin(omega*t_num)

    alpha1_dot = -a1_cos*omega*np.sin(omega*t_num) + a1_sin*omega*np.cos(omega*t_num)
    alpha2_dot = -a2_cos*omega*np.sin(omega*t_num) + a2_sin*omega*np.cos(omega*t_num)

    return alpha1, alpha2, alpha1_dot, alpha2_dot

# Swimmer dynamics
def swimmer_dynamics_fourier_offset(t_num, z, c):
    x_num, y_num, theta_num = z

    alpha1, alpha2, alpha1_dot, alpha2_dot = fourier_gait_with_offset(t_num, c)

    q_num = np.array([
        x_num,
        y_num,
        theta_num,
        alpha1,
        alpha2
    ], dtype=float)

    q_dot_num = np.array([
        0.0,
        0.0,
        0.0,
        alpha1_dot,
        alpha2_dot
    ], dtype=float)

    A_tilde_num = np.array(
        A_tilde_lambda(t_num, q_num, q_dot_num, params_num),
        dtype=float
    )

    shape_dot = np.array([
        alpha1_dot,
        alpha2_dot
    ], dtype=float)

    pose_dot = A_tilde_num @ shape_dot

```

```

    return pose_dot.flatten()

# Fast Euler simulation over one gait cycle
def simulate_one_cycle(c, dt=1/60):
    z = z0.copy()

    time_hist = np.arange(0.0, 1.0 + dt, dt)

    x_hist = []
    y_hist = []
    theta_hist = []

    alpha1_hist = []
    alpha2_hist = []
    alpha1dot_hist = []
    alpha2dot_hist = []

    xdot_hist = []
    ydot_hist = []
    thetadot_hist = []

    for ti in time_hist:
        alpha1, alpha2, alpha1_dot, alpha2_dot = fourier_gait_with_offset(ti, c)

        zdot = swimmer_dynamics_fourier_offset(ti, z, c)

        x_hist.append(z[0])
        y_hist.append(z[1])
        theta_hist.append(z[2])

        alpha1_hist.append(alpha1)
        alpha2_hist.append(alpha2)
        alpha1dot_hist.append(alpha1_dot)
        alpha2dot_hist.append(alpha2_dot)

        xdot_hist.append(zdot[0])
        ydot_hist.append(zdot[1])
        thetadot_hist.append(zdot[2])

        z = z + dt * zdot

        if not np.all(np.isfinite(z)):
            break

    n = len(x_hist)

    return {
        "time": np.array(time_hist[:n]),
        "x": np.array(x_hist),
        "y": np.array(y_hist),
        "theta": np.array(theta_hist),
        "alpha1": np.array(alpha1_hist),
        "alpha2": np.array(alpha2_hist),
        "alpha1_dot": np.array(alpha1dot_hist),
        "alpha2_dot": np.array(alpha2dot_hist),
    }

```

```

        "x_dot": np.array(xdot_hist),
        "y_dot": np.array(ydot_hist),
        "theta_dot": np.array(thetadot_hist),
    }

# COT = effort / net forward distance
def objective_min_COT(c):
    try:
        sim = simulate_one_cycle(c, dt=dt)

        if len(sim["x"]) < 5:
            return 1e6

        net_x = sim["x"][-1] - sim["x"][0]

        # Reject backward or tiny forward motion immediately
        if net_x <= 0.01:
            return 1e6

        alpha1_dot = sim["alpha1_dot"]
        alpha2_dot = sim["alpha2_dot"]

        # Faster effort approximation
        effort = dt * np.sum(alpha1_dot**2 + alpha2_dot**2)

        COT = effort / net_x

        if not np.isfinite(COT):
            return 1e6

        # Reject very inefficient gaits
        if COT > 200:
            return 1e6

        return COT

    except Exception:
        return 1e6

# Initial guess
try:
    c0_COT = c_opt_forward.copy()
except NameError:
    c0_COT = np.array([
        0.6,
        0.6,
        0.0,
        -np.pi/6,
        np.pi/6,
        0.0
    ], dtype=float)

# Bounds
bounds_COT = [
    (0.4, 0.9),      # alpha1 offset
    (0.4, 0.9),      # alpha2 offset

```

```

    (-0.7, 0.7),      # alpha1 cos
    (-0.7, 0.7),      # alpha1 sin
    (-0.7, 0.7),      # alpha2 cos
    (-0.7, 0.7),      # alpha2 sin
]

# Make sure c0 is inside bounds
c0_COT = np.array([
    np.clip(c0_COT[i], bounds_COT[i][0], bounds_COT[i][1])
    for i in range(len(c0_COT))
], dtype=float)

# Run COT optimization
result_COT = minimize(
    objective_min_COT,
    c0_COT,
    method="SLSQP",
    bounds=bounds_COT,
    options={
        "maxiter": 10,
        "ftol": 1e-5,
        "disp": True
    }
)

c_opt_COT = result_COT.x

print("COT optimization success:", result_COT.success)
print("Optimizer message:", result_COT.message)
print("Minimum COT:", result_COT.fun)
print("Optimized COT coefficients:")
print(c_opt_COT)

# Simulate optimized COT gait
sim_COT = simulate_one_cycle(c_opt_COT, dt=dt)

time = sim_COT["time"]
x_sol = sim_COT["x"]
y_sol = sim_COT["y"]
theta_sol = sim_COT["theta"]

alpha_1_sol = sim_COT["alpha1"]
alpha_2_sol = sim_COT["alpha2"]

alpha_1_dot_sol = sim_COT["alpha1_dot"]
alpha_2_dot_sol = sim_COT["alpha2_dot"]

x_dot_sol = sim_COT["x_dot"]
y_dot_sol = sim_COT["y_dot"]
theta_dot_sol = sim_COT["theta_dot"]

net_x = x_sol[-1] - x_sol[0]
net_y = y_sol[-1] - y_sol[0]
net_theta = theta_sol[-1] - theta_sol[0]

effort = dt * np.sum(alpha_1_dot_sol**2 + alpha_2_dot_sol**2)

```



```

print("Net x displacement:", net_x)
print("Net y displacement:", net_y)
print("Net theta change:", net_theta)
print("Effort:", effort)

if net_x <= 1e-5:
    print("Invalid COT: gait did not move forward.")
else:
    COT = effort / net_x
    print("Cost of Transport:", COT)

# Plot states and velocities
fig = plt.figure(4, figsize=(11, 10))
fig.clf()
ax = fig.subplots(5, 2, sharex=True)

ax[0][0].plot(time, x_sol)
ax[0][0].set_ylabel('x')

ax[1][0].plot(time, y_sol)
ax[1][0].set_ylabel('y')

ax[2][0].plot(time, theta_sol)
ax[2][0].set_ylabel('theta')

ax[3][0].plot(time, alpha_1_sol)
ax[3][0].set_ylabel('alpha_1')

ax[4][0].plot(time, alpha_2_sol)
ax[4][0].set_ylabel('alpha_2')
ax[4][0].set_xlabel('time')

ax[0][1].plot(time, x_dot_sol)
ax[0][1].set_ylabel('x_dot')

ax[1][1].plot(time, y_dot_sol)
ax[1][1].set_ylabel('y_dot')

ax[2][1].plot(time, theta_dot_sol)
ax[2][1].set_ylabel('theta_dot')

ax[3][1].plot(time, alpha_1_dot_sol)
ax[3][1].set_ylabel('alpha_dot_1')

ax[4][1].plot(time, alpha_2_dot_sol)
ax[4][1].set_ylabel('alpha_dot_2')
ax[4][1].set_xlabel('time')

plt.tight_layout()

# Shape-space plot
plt.figure(5, figsize=(5, 5))
plt.plot(alpha_1_sol, alpha_2_sol)
plt.scatter(alpha_1_sol[0], alpha_2_sol[0], label='start')
plt.scatter(alpha_1_sol[-1], alpha_2_sol[-1], label='end')

```

```

plt.xlabel(r'$\alpha_1$')
plt.ylabel(r'$\alpha_2$')
plt.title('Minimum COT gait in shape space')
plt.grid(True)
plt.axis('equal')
plt.legend()

# XY trajectory plot
plt.figure(6, figsize=(6, 4))
plt.plot(x_sol, y_sol)
plt.scatter(x_sol[0], y_sol[0], label='start')
plt.scatter(x_sol[-1], y_sol[-1], label='end')
plt.xlabel('x')
plt.ylabel('y')
plt.title('Minimum COT swimmer trajectory')
plt.grid(True)
plt.axis('equal')
plt.legend()

plt.show()

```

Optimization terminated successfully (Exit mode 0)

Current function value: 1000000.0

Iterations: 1

Function evaluations: 7

Gradient evaluations: 1

COT optimization success: True

Optimizer message: Optimization terminated successfully

Minimum COT: 1000000.0

Optimized COT coefficients:

[0.4 0.4 0.001185 0.7 0.7 -0.18419131]

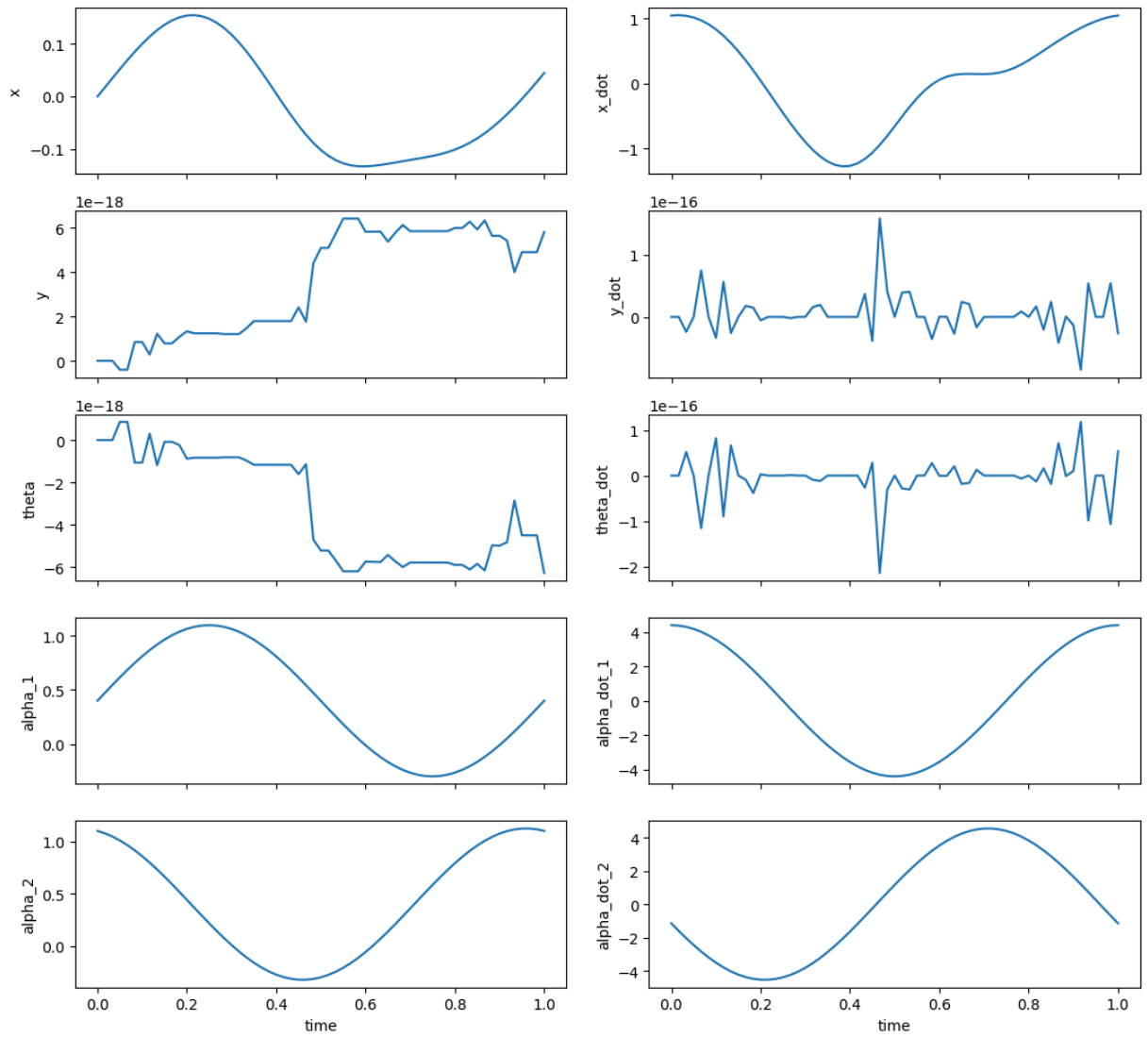
Net x displacement: 0.04477727784824301

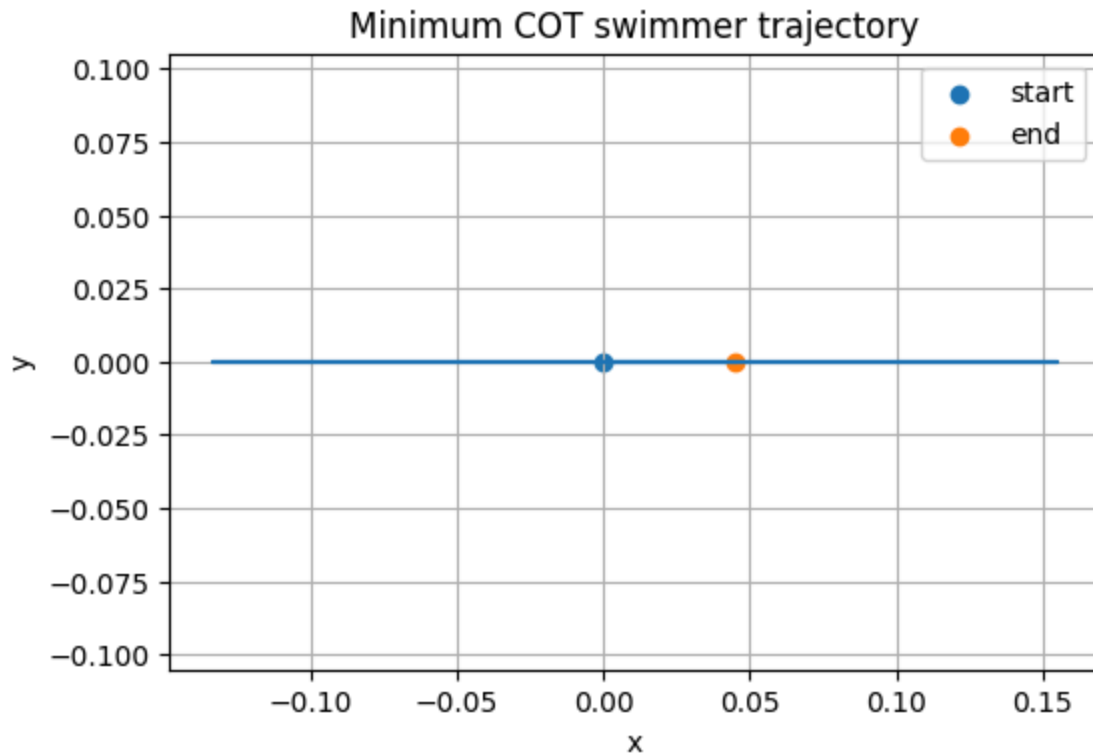
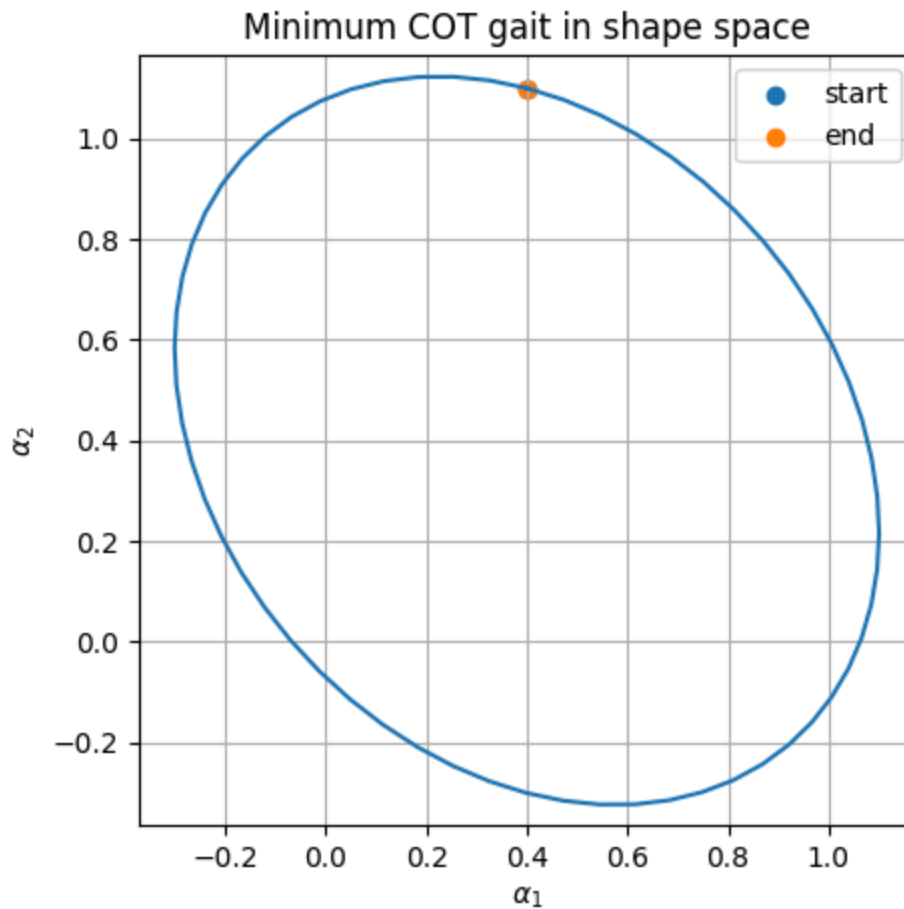
Net y displacement: 5.804801516727388e-18

Net theta change: -6.2919274286300276e-18

Effort: 20.35886320313229

Cost of Transport: 454.66951501901406





The optimized cost-of-transport gait produced a relatively large COT value, meaning the swimmer joint effort is very large relative to the achieved forward displacement. This

suggests the gait is not highly energy efficient, even though it still generates forward motion.

Please provide a short paragraph describing your thoughts on the required reading below. This can be in any form: tell me what you liked, didn't like, understood, didn't understand. There is no wrong answer as long as you read and reflect.

- Diaz, K., Robinson, T. L., Aydin, Y. O., Aydin, E., Goldman, D. I., & Wan, K. Y. (2021). A minimal robophysical model of quadriflagellate self-propulsion. *Bioinspiration & Biomimetics*, 16(6), 066001.

I found this paper really interesting because it connects motion at a microscale with robotics in order to see what capabilities could be used in robotics from nature. The author showed very easily that simply changing the coordination (gait) of the flagella, without changing the structure, can significantly affect swimming performance. It was easy to see how the trot gait was consistently the fastest in both real algae and the robot made the results feel that this gait was the most efficient out of all of them. I like how the paper also ties back concepts from lecture like low Reynolds number and the need to break symmetry change to achieve motion. One thing I found confusing was understanding how the phase differences between the flagella translate into differences in net displacement, but the figures helped to see this more clearly. Overall, I like the use of figures and structure of the paper. It was very well written and it was easy to follow. Using an actual robot as the physical model was a really effective approach because it allowed us to test various ideas that would be difficult to isolate in real organisms.